CIFAR-10

Two-Track Classification Study

Transfer Learning & Build From Scratch | What the Gap Reveals

Agenda

Zachary · CIFAR-10 CV Project · May 2026

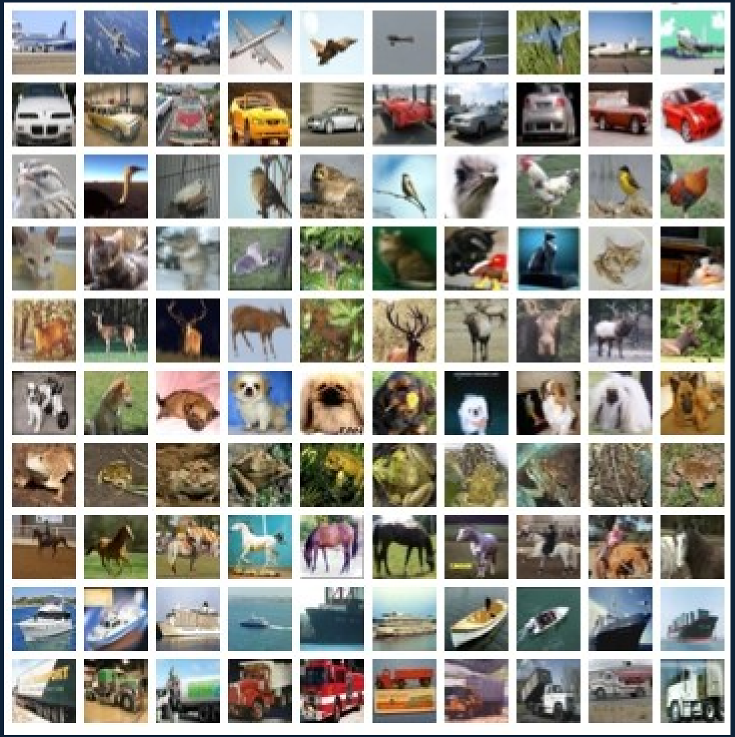
01 Computer Vision Classification

Problem

- 10-class image classification
- 32×32 pixel inputs — extreme resolution constraint
- Visual ambiguity: cat vs. dog vs. deer at low-res
- Upscaling required for pretrained architectures (32→224px)
- Shared goal: understand WHAT drives performance

Shared Methodology

- PyTorch – run local
- CrossEntropyLoss (softmax built-in)
- 80/20 train/val + 10k test
- Adam & AdamW: lr=1e-3,4,5
- Early stopping @ Patience = 3



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Approach

Track 1 — Transfer Learning

Size: 10K Train
ResNet50 + ImageNet weights
Frozen base → progressive unfreezing
32px upscaled to 224px
ImageNet normalization
Batch size=32

VS

Track 2 — From Scratch

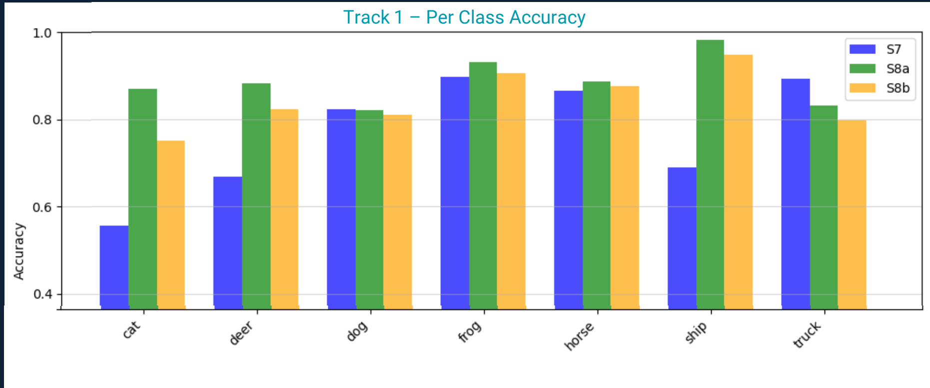
Size: Progressive 10K → 35K → 50K
Custom ResNet variant, Kaiming init
Native 32×32 — no upscaling
No pretrained priors
CIFAR-10 normalization

02 Track 1 — Transfer Learning

		Experimental Progression	Val Acc	Delta	Test Acc
Adam **lr=1e-3** | Epochs=10/8 Overfit **> 5** | Train Acc= **77.04%**	**S7**	Head only Baseline	80.85%	-2.83	**78.0%**
AdamW **lr=1e-4** | Epochs=10/4 Overfit **> 1** | Train Acc= **93.74%**	**S8a**	Full unfreeze	90.30%	-0.81	**89.5%**
AdamW **lr=1e-4** | Epochs=10/2 Overfit **> 1** | Train Acc= **93.11%**	**S8b**	Layer4 only	86.40%	-0.78	**85.6%**
AdamW **lr=1e-4** | Epochs=10/2 | Size=35k MIN_DELTA **> 0.005 → 0.01** | Train Acc= **95.66%**	**S11**	Progression Layer3→4only	92.89%	-0,04	**92.9%**

Transforms: Resize 32 → 224 | ImageNet norm | No augmentation
Flatten→ Linear128 → ReLU → Linear64 → ReLU → Dropout 0.2

conv1	Raw edges & color gradients
layer1	Simple textures & shapes
layer2	Patterns & object parts
layer3	Complex part combinations
layer4	High-level class features
fc head	Custom classifier



03 Track 2 — Build From Scratch

Key Architectural Decisions

- 3×3 stem (vs. 7×7)
- No MaxPool — preserve 32px spatial context
- Bottleneck blocks [2,2,2,2] — lighter depth
- Kaiming (He) init — calibrated for ReLU
- Dropout 0.1 per block — no pretrained priors to regularize
- CIFAR-10 native normalization
- Augmentation: RandomFlip + RandomCrop

Surprising findings →

- Breakthroughs with longer consolidation
- Allow more epochs
- min_delta critical
- Leverage Early Stopping
- Cats are lazy learners

Phased Dataset Progression

Phase 1 — 10k | Adam | lr=1e-3 | min_delta 0,01
5 classes at 0% — class collapse
25.2%

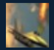
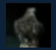
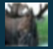
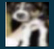
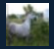
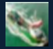
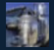
Phase 2 — 35k | AdamW | lr=1e-4 | min_delta 0,005
Cat & deer still 0% — needs scale
49.2%

Phase 3a — 50k | AdamW | lr=1e-5 | min_delta 0,005
Collapse resolves — cat reaches 27%
75.4%

Phase 3b
Cat 63.6% · Early stop @ epoch 14
81.2%

04 What the Comparison Reveals

11.7 % gap = **92.9%** *S11-R3 | **81.2%** *P-3b

Class	Track 1 Best	Track 2 Best
 airplane	95.7%	89%
 automobile	96.6%	95.3%
 bird	90.4%	77.5%
 **cat**	86.0%	63.6%
 **deer**	93.6%	72.7%
 **dog**	87.5%	68.4%
 frog	96.2%	82.5%
 horse	91.3%	88%
 ship	96.2%	90%
 truck	95.0%	85%

Gradient Monitoring — stem is doing the heavy lifting

Epoch	stem	layer1	layer2	layer3	layer4
01	**0.0619**	0.0082	0.0033	0.0017	0.0011
09	**0.0755**	0.0082	0.0032	0.0017	0.0010
14	**0.0461**	0.0051	0.0022	0.0013	0.0006

Stem 8–10× higher than layer1. | Layer4 stabilized earliest
Mirrors Track 1 layer selection finding.

Track 1				Track 2			
S7	S8a	S8b	S11	P1	P2	P3a	P3b
78.0	89.5	85.6	92.9	25.2	49.2	75.4	81.2

To be Continued.....

05 Weiter, weiter!

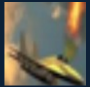
01 Gradient monitoring — rerun both tracks
Add gradient monitoring outputs to all training loops in both Track 1 and Track 2.
Which layers were most active, and does it change using upscaled 224px vs. native 32px?

02 Gradient-informed selective retraining
Track 1 — Progressive unfreezing layer3 then layer3+4 → using full 50k.
Track 2 — Unfreeze stem + layer1 using full 50k; previously showed highest gradient.

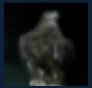
03 Batch size as a variable
Observe differences in layer learning rates across batch sizes.
Does batch size influence which layers adapt fastest?

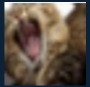
04 Open question
Can gradient monitoring inform transfer learning as the freeze-boundary signal
VS pretrained assumptions?

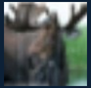
05 Ensemble Potential — Two Tracks, One Model?
Will combining models (Averaged Predictions or a lightweight Meta-model)
outperform, specifically with: Cat, deer, and dog?.

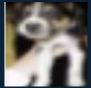

Airplane

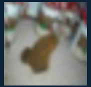

Automobile

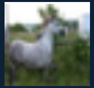

Bird

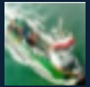

Cat

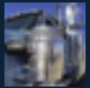

Deer


Dog


Frog


Horse

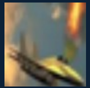

Ship


Truck

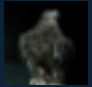
Track 1 Per-Class Accuracy -- Full Comparison

Class	S7 Head 10k	S10 Head 50k	S8a Full 10k	S11-R1 L3 35k	S11-R2 L4 35k	S11-R3 Stacked 35k
airplane	~89%	90.10%	~96%	95.20%	93.80%	95.70%
automobile	~91%	83.70%	~97%	95.20%	91.80%	96.60%
bird	~75%	83.00%	~92%	89.80%	86.80%	90.40%
cat	55.70%	69.70%	~82%	83.40%	85.00%	86.00%
deer	~68%	75.30%	~91%	92.20%	88.10%	93.60%
dog	~67%	74.70%	~88%	90.10%	82.90%	87.50%
frog	~90%	82.30%	~95%	96.70%	92.50%	96.20%
horse	~86%	85.00%	~96%	94.50%	94.10%	91.30%
ship	~91%	83.20%	~98%	96.10%	94.10%	96.20%
truck	~89%	90.80%	~96%	95.50%	94.80%	95.00%
Overall	**78.02%**	**81.78%**	**89.49%**	**92.87%**	**90.39%**	**92.85%**

Track 2 Per-Class Accuracy -- Full Comparison


Airplane


Automobile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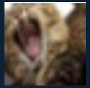
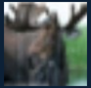

Cat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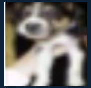
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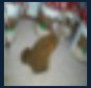
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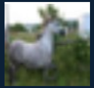
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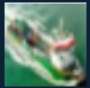
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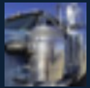
Truck

Phase	Dataset	Epochs	Test Acc	Cat	Deer	Dog	Horse
Phase 1	10k	20	25.15%	0%	0%	89.9%	0.4%
Phase 2	35k	20	49.15%	0%	0%	84.7%	0.4%
Phase 3a	50k	20	75.43%	27.2%	58.6%	75.4%	84.6%
Phase 3b	50k	14*	81.17%	63.6%	72.7%	68.4%	88.0%

Class	Correct	Total	Acc
airplane	890	1000	89.00%
automobile	953	1000	95.30%
bird	772	1000	77.20%
cat	636	1000	63.60%
deer	727	1000	72.70%
dog	684	1000	68.40%
frog	825	1000	82.50%
horse	880	1000	88.00%
ship	900	1000	90.00%
truck	850	1000	85.00%